

SECTION 6 – PRE SEA TRAINING AND CLASS

41. At which location on Earth does the Coriolis effect reach its greatest strength?
- A. At the Equator
 - B. At the Tropics
 - C. At 30° latitude
 - D. At the Poles
42. In the Northern Hemisphere, what is the typical direction of wind circulation around a low-pressure area?
- A. Anticlockwise and inward
 - B. Clockwise and inward
 - C. Clockwise and outward
 - D. Anticlockwise and outward
43. At what temperature does the air reach saturation, causing water vapor to begin condensing into liquid droplets?
- A. Temperature at which air becomes saturated
 - B. Temperature of condensation clouds
 - C. Lowest night temperature
 - D. Temperature at which rain begins
44. Which type of weather front is most commonly linked to severe thunderstorms, heavy rain, and sudden gusty winds?
- A. Cold front
 - B. Occluded front
 - C. Warm front
 - D. Stationary front
45. Which frontal type is MOST commonly associated with the process shown in the diagram?
- A. Stationary front
 - B. Warm front
 - C. Cold front
 - D. Occluded front
46. In radar navigation, what is the main purpose of using a parallel index line?
- A. Determining gyro error
 - B. Maintaining a constant distance from hazards or the coastline
 - C. Plotting tidal set and drift
 - D. Calculating the vessel's speed

47. On a nautical chart, which color is commonly used to indicate shallow water and the boundaries of certain regulated areas?

- A. Gold
- B. Blue
- C. Black
- D. Green

48. The vertical distance between the upper edge of the deck line and the centre of the load line disc represents the:

- A. Summer Freeboard
- B. Summer Draft
- C. Fresh Water Allowance
- D. Winter North Atlantic Freeboard

49. A vessel departs Felixstowe, England at 1000 hrs local time (GMT+0) on 2nd March 2014, bound for Kolkata, India, a distance of 10,080 nautical miles. If the vessel maintains a constant speed of 10 knots, what will be the vessel's estimated time of arrival (ETA) at Kolkata in local time (GMT+5.5)?

- A. 13th April 2014, 1000 hrs LT
- B. 13th April 2014, 0430 hrs LT
- C. 13th April 2014, 1530 hrs LT
- D. 13th April 2014, 1200 hrs LT

50. The azimuth of a celestial body is measured at which of the following locations?

- A. Observer's Zenith
- B. Observer's Nadir
- C. Observer's Meridian
- D. Observer's Prime Vertical

51. Which method is most commonly used to determine compass error during sunrise or sunset?

- A. Latitude by meridian passage
- B. Azimuth
- C. Sextant altitude
- D. Amplitude

52. In celestial navigation, what is the First Point of Aries (♈) primarily used to measure?

- A. Measuring True Altitude

- B. Measuring Azimuth
- C. Measuring Right Ascension
- D. Measuring Longitude

53. Which of the following corrections must be applied to a sextant reading to accurately determine the observed altitude (H_o) of a celestial body?

- A. Index error, dip, refraction, semi-diameter & parallax
- B. Index error and dip only
- C. Refraction only
- D. Dip and refraction only

54. What is the primary navigational purpose of observing the Sun at its meridian passage?

- A. Applying time correction
- B. Finding longitude
- C. Calculating compass error
- D. Determining latitude

55. In celestial navigation, what is the reason for drawing the intercept at right angles to the azimuth direction?

- A. Because the azimuth determines the observer's latitude
- B. Because the Line of Position (LOP) is always perpendicular to the azimuth to the celestial body
- C. Because the intercept is affected by time zones
- D. Because the Earth's magnetic field influences the plotting

56. The observer's prime vertical passes through the east and west points of which of the following?

- A. Rational horizon
- B. Equinoctial
- C. Celestial meridian
- D. Great circle

57. The total area beneath the GZ (righting arm) curve from 0° to 40° of heel is a measure of which of the following vessel characteristics?

- A. The vessel's dynamic stability within that heel range
- B. The vessel's metacentric height (GM)
- C. The vessel's KM
- D. The vessel's maximum righting lever

58. According to ship safety regulations, which of the following must be made weathertight as part of the Conditions of Assignment?

- A. Hatchways situated on the freeboard deck
- B. Sounding pipes on deck
- C. Cargo hold ventilators below deck
- D. Scuppers above deck

59. Which opening below is NOT considered while assessing Conditions of Assignment?

- A. Hatchway on freeboard deck
- B. Door in superstructure end bulkhead
- C. Sounding pipe opening in engine room
- D. Ventilator on freeboard deck

60. What does the term "angle of loll" describe in ship stability?

- A. The angle at which the center of gravity is vertically above the center of buoyancy
- B. The time taken for a vessel to complete one roll cycle
- C. KM is greater than GM
- D. The heel angle at which a vessel with negative GM achieves equilibrium due to hull shape

61. Ventilators forming part of Conditions of Assignment must:

- A. Be of any height if fitted with wire mesh
- B. Have minimum height and efficient closing appliances
- C. Have closing devices only above superstructure decks
- D. Be permanently open for machinery ventilation

62. A navigational aid emits light for a shorter time than it remains dark in each cycle. What is this light called?

- A. Isophase
- B. Occulting
- C. Intermittent
- D. Flashing

63. When approaching a danger marked by a Cardinal buoy, the vessel should:

- A. Pass on the port side
- B. Pass on the starboard side
- C. Keep to the side named by the buoy
- D. Keep the buoy on the vessel's port side

64. How should a newly issued Preliminary Notice to Mariners (P-Notice) be marked on a nautical chart?

- A. In red marker

- B. Lightly in pencil
- C. In permanent ink
- D. Not marked until confirmed

65. When crossing a traffic separation scheme, the vessel should:

- A. Cross at any convenient angle
- B. Cross at right angles to the traffic flow
- C. Alter course to avoid lanes
- D. Follow the traffic flow

66. A ship has a mean draft of 9.10 meters and a TPC (Tonnes per Centimeter) of 200. If 1800 metric tonnes of cargo are loaded, what will be the vessel's new mean draft?

- A. 10.10 m
- B. 9.19 m
- C. 9.80 m
- D. 9.99 m

67. Which factor causes the settling error of a gyrocompass to become more pronounced?

- A. Operating at higher latitudes
- B. Operating at lower latitudes
- C. Increasing the vessel's speed
- D. Navigating in areas of high magnetic variation

68. Aneroid barometers are typically calibrated to display atmospheric pressure in which of the following units?

- A. Inches of mercury and millibars
- B. Feet of mercury and millibars
- C. Inches of mercury and millimeters
- D. Inches of mercury and centimeters

69. Which instrument is specifically used to measure the apparent wind direction and speed on a moving ship?

- A. Barometer
- B. Psychrometer
- C. Hygrometer
- D. Anemometer

70. On which radar display mode does the own ship remain stationary at the center while other vessels and objects appear to move around it?

- A. North Up
- B. Course Up
- C. Relative Motion
- D. True Motion

71. Which celestial body is commonly used for obtaining a compass error by azimuth during twilight?

- A. Polaris
- B. Venus
- C. The Sun
- D. Mars

72. In marine meteorology, what does a rapidly falling barometer usually indicate?

- A. Stable weather conditions
- B. Approaching bad weather or storm
- C. Extremely dry air
- D. Calm sea conditions

73. Which cloud type is most commonly associated with thunderstorms and heavy rainfall?

- A. Cirrus
- B. Stratus
- C. Cumulonimbus
- D. Altocumulus

74. What is the primary purpose of a ship's freeboard?

- A. To increase cargo capacity
- B. To improve engine efficiency
- C. To provide reserve buoyancy and safety
- D. To reduce fuel consumption

75. In navigation, what does "dead reckoning" primarily rely upon?

- A. Radar observations
- B. Celestial sightings
- C. Estimated course and speed
- D. Satellite communication

76. What is the SI unit used to measure atmospheric pressure?

- A. Knot
- B. Pascal
- C. Joule
- D. Newton

77. Which part of a ship experiences the greatest bending moment in heavy seas?

- A. Midship section
- B. Forecastle
- C. Stern peak
- D. Bridge deck

78. Which instrument is used to measure the salinity of seawater?

- A. Hygrometer
- B. Sextant
- C. Salinometer
- D. Barograph

79. What is the purpose of ballast water in a ship?

- A. To clean cargo tanks
- B. To improve ship stability and trim
- C. To increase engine efficiency
- D. To cool cargo holds

80. What does the term "lee shore" refer to in seamanship?

- A. A shore protected from wind
- B. A shore facing the wind
- C. A dangerous shore onto which wind blows
- D. A shallow coastal area

81. Which navigation light is carried by a vessel not under command?

- A. Two all-round red lights in a vertical line
- B. One all-round white light
- C. Green and red sidelights only
- D. Two all-round green lights

82. In ship stability, GM refers to:

- A. Gross Measurement
- B. General Metric
- C. Metacentric Height

D. Gravity Measurement

83. What is the primary purpose of a marine sextant?

- A. Measuring wind direction
- B. Determining celestial altitude
- C. Measuring sea depth
- D. Determining engine power

84. Which factor most affects the rolling period of a ship?

- A. Cargo color
- B. Metacentric height
- C. Number of lifeboats
- D. Radar frequency

85. Which chart projection is most commonly used for marine navigation?

- A. Gnomonic Projection
- B. Mercator Projection
- C. Polar Projection
- D. Cylindrical Equal Area Projection

86. A vessel alters course from 090° to 180°. What is the amount of alteration?

- A. 45°
- B. 90°
- C. 180°
- D. 270°

87. Which emergency signal indicates abandoning ship?

- A. Continuous ringing of bell
- B. Seven short blasts followed by one long blast
- C. Three prolonged blasts
- D. Morse code SOS only

88. Which gas is primarily responsible for the greenhouse effect?

- A. Oxygen
- B. Nitrogen
- C. Carbon dioxide
- D. Helium

89. What is the function of a ship's rudder?

- A. To increase stability
- B. To change the ship's direction
- C. To reduce rolling
- D. To measure speed

90. Which side of a vessel is indicated by a green navigation light?

- A. Port side
- B. Starboard side
- C. Stern
- D. Masthead

91. What is the main purpose of a ship's anchor?

- A. To increase vessel speed
- B. To hold the vessel in position
- C. To improve fuel efficiency
- D. To reduce rolling motion

92. Which instrument is used to determine the depth of water beneath a vessel?

- A. Barometer
- B. Echo Sounder
- C. Sextant
- D. Hygrometer

93. What does SOLAS stand for?

- A. Safety of Life at Sea
- B. Security of Life and Ships
- C. Standard Operations for Large Ships
- D. Safety Operations and Life Saving

94. Which type of fire extinguisher is best suited for electrical fires?

- A. Water extinguisher
- B. Foam extinguisher
- C. CO₂ extinguisher
- D. Dry sand

95. What is the standard distress frequency used in marine VHF communication?

- A. Channel 6

- B. Channel 13
- C. Channel 16
- D. Channel 21

96. Which part of the ship is located at the front?

- A. Stern
- B. Bow
- C. Port
- D. Starboard

97. What does "starboard" refer to on a ship?

- A. Left side
- B. Front side
- C. Right side
- D. Rear side

98. Which safety equipment is used to prevent drowning during emergencies?

- A. Compass
- B. Lifebuoy
- C. Sextant
- D. Radar reflector

99. What is the purpose of a ship's compass?

- A. To measure sea depth
- B. To determine direction
- C. To measure wind speed
- D. To detect underwater objects

100. Which international organization regulates global shipping standards?

- A. NASA
- B. WHO
- C. IMO
- D. UNESCO

101. Which emergency equipment is used to send distress signals to nearby ships?

- A. EPIRB
- B. Gyro Compass
- C. Echo Sounder
- D. Barograph

102. What does "port side" mean on a vessel?

- A. Right side
- B. Left side
- C. Front side
- D. Upper deck

103. What is the main function of a life raft?

- A. Cargo storage
- B. Emergency accommodation after abandoning ship
- C. Firefighting
- D. Navigation support

104. Which document contains detailed information about cargo carried on a vessel?

- A. Deck Log Book
- B. Cargo Manifest
- C. Engine Log Book
- D. Stability Booklet

105. What is the purpose of a ship's whistle or horn?

- A. Entertainment
- B. Distress communication only
- C. Navigational sound signaling
- D. Engine warning

106. What does "abandon ship" mean?

- A. Leaving cargo operations
- B. Evacuating the vessel during emergency
- C. Docking at a port
- D. Entering restricted waters

107. Which factor mainly affects ocean tides?

- A. Wind speed
- B. Earthquake activity
- C. Gravitational pull of the Moon and Sun
- D. Atmospheric pressure only

108. What is the full form of GPS?

- A. Global Positioning System
- B. General Port System
- C. Global Port Signal
- D. Geographic Position Source

109. Which type of knot is commonly used to secure a rope to a post?

- A. Reef Knot
- B. Bowline
- C. Clove Hitch
- D. Sheet Bend

110. What is the primary purpose of a ship's radar?

- A. Cooking assistance
- B. Detecting objects and navigation
- C. Measuring cargo weight
- D. Monitoring fuel usage

111. What is the full form of CSK?

- A. Chennai Super Kings
- B. DDDD
- C. SSSS
- D. FFFF